Exam 1 — Question 2

Question. Apply gradient descent to

$$\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The matrix has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = -2$, with eigenvectors $v_1 = \frac{1}{\sqrt{2}}(1,1)$ and $v_2 = \frac{1}{\sqrt{2}}(1,-1)$. The unique solution is $x^\star = (1/4, 1/4)$.

(a) What is the maximum rate $\rho^\star$ below which convergence is guaranteed for every initialization?

Answer. Let A denote the matrix and use the sum-of-squares loss

$$F(z) = \|Az - b\|_2^2, \qquad \nabla F(z) = 2A^\mathsf{T}(Az - b).$$

If $e_t = z_t - z^\star$, then one gradient-descent step gives

$$e_{t+1} = (I - 2\rho A^\mathsf{T} A)e_t.$$

Since A is symmetric, an eigenvalue λ of A produces eigenvalue λ^2 for $A^\mathsf{T}A$. The corresponding error multiplier is $1 - 2\rho\lambda^2$. Convergence for every initialization requires

$$|1 - 2\rho\lambda^2| < 1 \quad \text{for both eigenvalues.}$$

The largest squared eigenvalue is $4^2 = 16$, so

$$0 < \rho < \frac{1}{16}, \qquad \boxed{\rho^\star = \frac{1}{16}}.$$

(b) With $\rho = 0.01$ and $z_0 = (1, 0)$, the distance to $z^\star$ is $\Theta(b^t)$. Find b.

Answer. The two error multipliers are

$$q_1 = 1 - 2(0.01)(4^2) = 0.68, \qquad q_2 = 1 - 2(0.01)(-2)^2 = 0.92.$$

The initial error has a nonzero component in the v_2 direction. Therefore the slower surviving factor controls the asymptotic distance:

$$\|z_t - z^\star\|_2 = \Theta(0.92^t), \qquad \boxed{b = 0.92}.$$

Exam 1 — Question 3

Question. Calculate the Boolean Fourier transforms below. The input order is $(++, -+, +-, --)$.

(a) $f = [1, 1, -1, -1]$.

Answer. This is exactly the function $f(x_1, x_2) = x_2$. Hence

$$\widehat{f}(\{2\}) = 1, \qquad \widehat{f}(S) = 0 \quad (S \neq \{2\}).$$

(b) $g = [0, 1, 1, 1]$.

Answer. The function $1 - g$ is the indicator of the point $(+,+)$:

$$1 - g = \text{point}_{++}(x_1, x_2) = \frac{(1 + x_1)(1 + x_2)}{4}.$$

Expanding and subtracting from 1 gives

$$\boxed{g = \frac{3}{4} - \frac{1}{4}x_1 - \frac{1}{4}x_2 - \frac{1}{4}x_1x_2}.$$

Thus its coefficients at $S = \varnothing, \{1\}, \{2\}, \{1,2\}$ are $3/4, -1/4, -1/4, -1/4$.

(c) Let

$$h(x_1, x_2, x_3) = \begin{cases} f(x_1, x_2), & x_3 = -1, \\ g(x_1, x_2), & x_3 = +1. \end{cases}$$

Answer. The factors $(1 - x_3)/2$ and $(1 + x_3)/2$ select the two cases:

$$h = \frac{1 - x_3}{2} f + \frac{1 + x_3}{2} g = \frac{f + g}{2} + x_3 \frac{g - f}{2}.$$

Substituting $f = x_2$ and the polynomial for g gives

$$\boxed{\begin{aligned} h = \frac{3}{8} - \frac{1}{8}x_1 + \frac{3}{8}x_2 - \frac{1}{8}x_1x_2 \\ + \frac{3}{8}x_3 - \frac{1}{8}x_1x_3 - \frac{5}{8}x_2x_3 - \frac{1}{8}x_1x_2x_3. \end{aligned}}$$

Therefore, in the order $\varnothing, 1, 2, 3, 12, 13, 23, 123$, the coefficient list is

$$\boxed{\frac{3}{8}, -\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, -\frac{1}{8}, -\frac{1}{8}, -\frac{5}{8}, -\frac{1}{8}}.$$

Exam 2 — Question 2

Question. Let A be the adjacency matrix of the 3-cycle, with zeros on the diagonal. Its largest eigenvalue is $\lambda_1 = 2$, with eigenvector $v_1 = \frac{1}{\sqrt{3}}(1,1,1)$.

(a) If $x \perp v_1$, prove that normalized power iteration oscillates between x and $-x$.

Answer. Write $x = (x_1, x_2, x_3)$. Orthogonality gives

$$x \cdot v_1 = 0 \quad \Longrightarrow \quad x_1 + x_2 + x_3 = 0.$$

Multiplication by the adjacency matrix yields

$$Ax = (x_2 + x_3,\ x_3 + x_1,\ x_1 + x_2).$$

Because the coordinate sum is zero, $x_2 + x_3 = -x_1$, $x_3 + x_1 = -x_2$, and $x_1 + x_2 = -x_3$. Therefore

$$Ax = (-x_1, -x_2, -x_3) = -x.$$

If x is normalized, then $\|Ax\| = \|-x\| = 1$. Thus one iteration maps x to $-x$ and the next maps $-x$ back to x.

$$\boxed{x, -x, x, -x, \dots}$$

(b) Deduce the other two eigenvalues.

Answer. The orthogonal complement of v_1 is two-dimensional, and the calculation above shows that A acts as $-I$ on every vector in that space. Hence

$$\boxed{\lambda_2 = \lambda_3 = -1}.$$

Exam 2 — Question 3

Question. At $x = 1, i, -1, -i$, the values of f are $0, 1, 2, 3$. Use

$$f(x) = f^+(x^2) + x f^-(x^2)$$

to recover the polynomial representation of f.

(a) Find the value lists of f^+ and f^-.

Answer. Evaluate the decomposition at x and $-x$. Adding eliminates the odd part and subtracting eliminates the even part:

$$f^+(x^2) = \frac{f(x) + f(-x)}{2}, \qquad f^-(x^2) = \frac{f(x) - f(-x)}{2x}.$$

At $x = 1$,

$$f^+(1) = \frac{0+2}{2} = 1, \qquad f^-(1) = \frac{0-2}{2} = -1.$$

At $x = i$, where $i^2 = -1$,

$$f^+(-1) = \frac{1+3}{2} = 2, \qquad f^-(-1) = \frac{1-3}{2i} = -\frac{1}{i} = i.$$

Thus, over the domain $(1, -1)$,

$$\boxed{f^+ = [1, 2]}, \qquad \boxed{f^- = [-1, i]}.$$

(b) Find the degree-one polynomial representations of f^+ and f^-.

Answer. If $p(y) = \alpha + \beta y$, $p(1) = u$, and $p(-1) = v$, then

$$\alpha = \frac{u+v}{2}, \qquad \beta = \frac{u-v}{2}.$$

Applying this formula to the two lists gives

$$f^+(y) = \frac{3}{2} - \frac{1}{2}y,$$

$$f^-(y) = \frac{-1+i}{2} + \frac{-1-i}{2}y.$$

(c) Reconstruct f.

Answer. Substitute $y = x^2$ and multiply the odd polynomial by x:

$$\boxed{f(x) = \frac{3}{2} + \frac{-1+i}{2}x - \frac{1}{2}x^2 + \frac{-1-i}{2}x^3}.$$

Exam 3 — Question 1

Question. Apply gradient descent to the inconsistent scalar system $x = 0$ and $x = 1$.

(a) Write the sum-of-squares loss and find its minimizer.

Answer. The residuals are x and $x - 1$, so

$$F(x) = x^2 + (x-1)^2.$$

Differentiate:

$$F'(x) = 2x + 2(x-1) = 4x - 2.$$

The minimum occurs where $F'(x^\star) = 0$, hence

$$\boxed{x^\star = \frac{1}{2}}.$$

(b) Find $\dfrac{x_t - x^\star}{x_{t-1} - x^\star}$.

Answer. One gradient-descent step is

$$x_t = x_{t-1} - \rho(4x_{t-1} - 2).$$

Subtract $x^\star = 1/2$ and factor:

$$x_t - \frac{1}{2} = (1 - 4\rho)\left(x_{t-1} - \frac{1}{2}\right).$$

Therefore

$$\boxed{\frac{x_t - x^\star}{x_{t-1} - x^\star} = 1 - 4\rho}.$$

(c) If $x_0 = 0$ and $\rho = 1/8$, find x_{20}.

Answer. The multiplier is $1 - 4(1/8) = 1/2$. Applying the recurrence twenty times gives

$$x_{20} - \frac{1}{2} = \left(\frac{1}{2}\right)^{20}\left(0 - \frac{1}{2}\right) = -\frac{1}{2^{21}}.$$

Thus

$$\boxed{x_{20} = \frac{1}{2} - \frac{1}{2^{21}}}.$$

Exam 3 — Question 2(a–b)

Question. Apply subspace iteration to

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix},$$

starting from the standard basis e_1, e_2, e_3. Ignore e_3 initially.

(a) Show the first three rounds for the first two vectors.

Answer. Define

$$u = \frac{1}{\sqrt{5}}(0, 1, 2).$$

Round 1.

$$Ae_1 = (0,1,2) = \sqrt{5}\,u, \qquad Ae_2 = (1,0,0) = e_1.$$

After normalization, $(x_1, x_2) = (u, e_1)$.

Round 2.

$$Au = (\sqrt{5}, 0, 0) = \sqrt{5}\,e_1, \qquad Ae_1 = \sqrt{5}\,u.$$

After normalization, $(x_1, x_2) = (e_1, u)$.

Round 3. The same computation switches them again, so $(x_1, x_2) = (u, e_1)$.

(b) Use this behavior to find two eigenvalues.

Answer. Since $Ae_1 = \sqrt{5}\,u$ and $Au = \sqrt{5}\,e_1$,

$$A(e_1 + u) = \sqrt{5}(e_1 + u),$$

$$A(e_1 - u) = -\sqrt{5}(e_1 - u).$$

Therefore

$$\boxed{\lambda_1 = \sqrt{5}, \qquad \lambda_2 = -\sqrt{5}}.$$

Exam 3 — Question 2(c–d)

Question. Prove that if Ax_i and Ax_j become linearly dependent for $i \neq j$, then zero is an eigenvalue. Use this to find the remaining eigenvalue of the matrix.

Answer. Linear dependence means that some scalar c satisfies $Ax_i = cAx_j$. By linearity,

$$A(x_i - cx_j) = Ax_i - cAx_j = 0.$$

The subspace-iteration state vectors are orthogonal, hence independent. Therefore $x_i - cx_j \neq 0$.

Conclusion. A sends a nonzero vector to zero, so that vector is an eigenvector with eigenvalue 0.

For the given matrix, dependence occurs in the first round:

$$Ae_2 = (1,0,0), \qquad Ae_3 = (2,0,0) = 2Ae_2.$$

Consequently,

$$A(2e_2 - e_3) = 0,$$

and the remaining eigenvalue is 0. The complete spectrum is

$$\boxed{\{\sqrt{5}, -\sqrt{5}, 0\}}.$$

Exam 3 — Question 3

Question. The two-bit addressing function is

$$\mathrm{addr}(x, y, z_{++}, z_{+-}, z_{-+}, z_{--}) = z_{xy}.$$

(a) Express it using the four point functions.

(b) Calculate $\widehat{\mathrm{addr}}(\varnothing)$, $\widehat{\mathrm{addr}}(\{1,2,3\})$, and $\widehat{\mathrm{addr}}(\{1,2,3,4\})$, where variables 3 and 4 are z_{++} and z_{+-}.

Answer. For $a, b \in \{-1, +1\}$, define

$$\mathrm{point}_{ab}(x, y) = \frac{(1 + ax)(1 + by)}{4}.$$

Exactly one point function equals 1 for each input (x, y), so

$$\boxed{\mathrm{addr} = \sum_{a,b \in \{\pm 1\}} z_{ab}\, \mathrm{point}_{ab}(x, y)}.$$

Equivalently, this is the sum of the $++, +-, -+, --$ terms.

Constant coefficient. Every summand contains one z variable, so no constant term appears:

$$\boxed{\widehat{\mathrm{addr}}(\varnothing) = 0}.$$

Coefficient indexed by $\{1,2,3\}$. This is the coefficient of xyz_{++}. Expand only the $++$ summand:

$$\begin{aligned} z_{++}\, \mathrm{point}_{++}(x, y) &= \frac{z_{++}(1 + x)(1 + y)}{4} \\ &= \frac{1}{4}\left(z_{++} + xz_{++} + yz_{++} + xyz_{++}\right). \end{aligned}$$

Thus

$$\boxed{\widehat{\mathrm{addr}}(\{1,2,3\}) = \frac{1}{4}}.$$

Coefficient indexed by $\{1,2,3,4\}$. This asks for the monomial $xyz_{++}z_{+-}$. Every summand contains exactly one z variable, never a product of two different z variables. Therefore

$$\boxed{\widehat{\mathrm{addr}}(\{1,2,3,4\}) = 0}.$$

Conclusion. Equivalently, every point function has degree 2 in x, y, and multiplying by one addressed bit gives degree at most 3. The degree-4 monomial $xyz_{++}z_{+-}$ cannot occur.